# SCIR: A Weakly Supervised Contextual Instance Refinement Method for Remote Sensing Object Detection

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***Abstract*—Most weakly supervised object detection (WSOD) methods currently prioritize the top-scoring object instance from proposals to train the corresponding object detector. The detector tends to focus on the entire object by analyzing the contextual information around the most discriminative activation regions. However, the traditional selective search (SS) algorithm yields top-scoring instance proposals that cover only a portion of the object, thereby diminishing the detector's sensitivity to objects with a wide range of scale variations in remote sensing images (RSIs), particularly tiny object clusters. To address this issue, this article proposes a novel WSOD method called the segment anything model (SAM)-guided proposal generation with contextual instance refinement (SCIR) method for detecting objects with significant scale variations in RSIs. Specifically, a SAM-guided proposal generator (SPG) module is designed to generate high-quality proposals based on SAM masks rather than the traditional top-scoring one. Our SPG combines WSOD's advantage of mining classification clues through inexact supervision with SAM's capability of prelearned world knowledge to provide automatic prompts for the SAM. Meanwhile, we propose a contextual enhanced feature extractor (CEFE) module to capture the global context of the visual scene, which further activates the feature representation of the entire object. Finally, the feature map from the CEFE and proposals generated by the SPG are fed into a context-perceived instance refinement (CPIR) module. Our CPIR aims to shift the attention of the detection network from the local feature portion to the entire object by integrating local and global contextual information. Extensive experiments on the challenging DOTA and DIOR datasets demonstrate that our proposed SCIR achieves state-of-the-art (SOTA) performance and is quite effective on multiscale object issues.**

***Index Terms*—Contextual information, remote sensing images (RSIs), segment anything model (SAM), weakly supervised object detection (WSOD).**

Received 2 September 2025; revised 1 November 2025; accepted 1 December 2025. Date of publication 4 December 2025; date of current version 11 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62372348, in part by the Key Research and Development Program of Shaanxi under Grant 2024GX-ZDCYL-02-10, and in part by Shaanxi Outstanding Youth Science Fund Project under Grant 2023-JCJQ-53. *(Corresponding author: Xi Yang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3640394

## I. Introduction

With the development of modern remote sensor technologies, object detection in remote sensing images (RSIs) has emerged as an essential task for numerous aerial and satellite image applications. As shown in Fig. 1(a), traditional object detection methods usually require fine bounding boxes to train models, but the labeling of bounding boxes is time-consuming and laborious, especially in RSIs [1], [2]. On the contrary, weakly supervised object detection (WSOD) has gained increasing attention owing to its requirement for only image-level annotations. As a significant research direction in the computer vision field nowadays, typical WSOD approaches [3], [4], [5], [6] in RSIs are mainly inspired by the object detection algorithms designed for natural scenes and aiming to handle the suboptimal issue. Despite their successes, these ill-posed solutions overlook the unique characteristics of RSIs. Different from natural images, remote sensing object detection still faces numerous difficulties and challenges, such as scale variations, small and dense objects, and complex background, as, respectively, shown in Fig. 1(b)–(d).

Most popular WSOD methods like [4], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23] are characterized by nonconvex optimization processes, which implies that the model requires multiple local optima influenced by initial proposals. Furthermore, three unique characteristics of RSIs critically exacerbate the difficulty in handling phenomena shown in Fig. 1(e) and (f) of these WSOD methods: 1) scale variations in Fig. 1(b) cause conventional proposal generators to yield low-quality proposals failing to cover size-varying objects, particularly tiny ones; 2) small and dense objects in Fig. 1(c) induce part ambiguities as WSOD methods misclassify fragments within clustered objects; and 3) complex backgrounds in Fig. 1(d) heighten over-reliance on top-scoring proposals, diverting detectors toward salient background patches or partial objects.

Thus, these key phenomena shown in Fig. 1(e) and (f) contribute to three challenges.

1) *Low-Quality Proposals for Object Detectors:* Conventional proposal generators exhibit weak extraction capabilities across various semantic levels because they rely solely on low-level features computed by superpixels [24] or bounding boxes [25].

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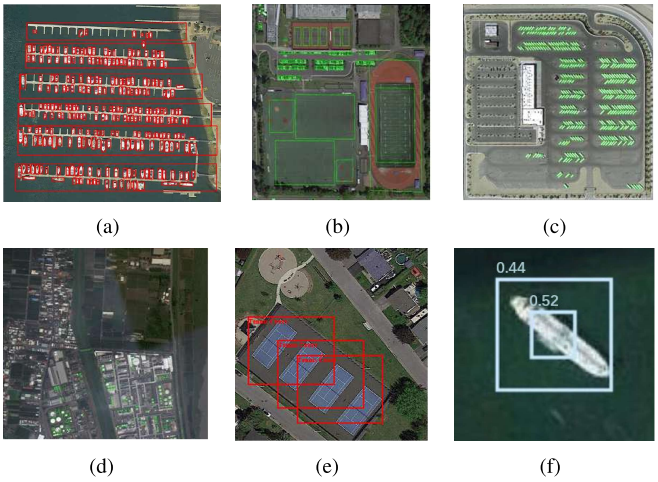



Fig. 1. Difficulties and phenomena of WSOD in RSIs. (a) Example of manual labeling on the remote sensing dataset. (b) Wide range of object scale variation. (c) Tiny, densely arranged objects. (d) High background complexity. (e) Dependence on conventional proposal generators and prioritization of discriminative parts rather than entire objects. (f) Selecting only the most confident and top-scoring proposals to train the detector.

2) *Potential Ambiguities for Different Parts of Objects:* To overcome the problem of "nonconvex optimization processes," some current popular WSOD methods [9], [10], [26] inspired by OICR [8] lead to potential ambiguities by simultaneously assigning the same label to distinct parts of objects.
3) *Top-Scoring Proposals:* These proposals often tend to cover only a portion of the object or even confused backgrounds, particularly in large-scale cluttered RSIs like Fig. 1(d). Furthermore, due to the RSIs' characteristics of the complex background and susceptibility of objects to occlusion, recent WSOD approaches still do not adequately address the above three challenges posed by the issue of "nonconvex optimization processes."

To solve the above WSOD challenges exacerbated by RSIs' characteristics, this article proposes a segment anything model (SAM)-guided proposal generation with contextual instance refinement (SCIR) method to perform WSOD in RSIs. Our SCIR consists of three modules: a SAM-guided proposal generator (SPG), a contextual enhanced feature extractor (CEFE), and context-perceived instance refinement (CPIR).

For the problem of low-quality proposals for object detectors, our SPG adopts classification clues as prompts from the SAM to autonomously generate proposals via multiple multi-aware prompt generation (MAPG) modules. These generated proposals are subsequently employed during WSOD training to promote class-conscious perception. The dependence of the SAM on manual hints and the lack of external knowledge of weakly supervised learning (WSL) drive our initiative to integrate the SAM into the WSL framework. The SPG framework is tailored to leverage the transferable knowledge embedded within the SAM, thus enhancing the WSL process.

To address the problem of potential ambiguities for different parts of objects, we meticulously devise a novel CEFE module to emphasize the holistic object features through a self-attention mechanism. Specifically, we first explicitly aggregate the features across all positions via a weighted average to generate global context features. Subsequently, the channel-wise interdependencies within the acquired global context are captured through an excitation operation. Ultimately, effective enhancement of instance-containing features is achieved by integrating global context features into those across all positions, thereby activating regions containing instances while suppressing background regions. Consequently, the module demonstrates efficacy in identifying high-quality instances to facilitate the training of more powerful object detectors.

To tackle the problem of top-scoring proposals, we intricately introduce an advanced end-to-end CPIR module. It aims at redirecting the attention of the object detection network away from local distinctive parts to the entire object, as well as potential additional instances. This is achieved by harnessing both local and global context information. Specifically, rather than solely selecting the highest-scoring proposal as pseudo-supervision, we initially identify a confident proposal among cluttered candidate boxes and delineate its surrounding regions. We subsequently aggregate local contextual information from these surrounding regions into a unified context representation. The adjusted confidence is finally derived based on this unified context representation.

Based on the above analysis, the main contributions of this article are as follows.

1) We present an SPG module with multiple MAPG that automatically guides the SAM using classification clues to generate proposals. The proposals produced by the SPG enhance both the effectiveness and efficiency of WSOD.
2) We design a novel CEFE module that effectively accentuates high-quality entire objects by leveraging the global context of a visual scene. Besides, the structure of the CEFE improved by the feature pyramid network (FPN) [27] effectively highlights feature information of tiny objects.
3) We propose a CPIR module that utilizes contextual information from surrounding regions to suppress low-quality parts of objects and emphasize high-quality whole objects. Moreover, we enhance regression supervision in the regression branch as a way to dynamically correct the object detection frame more precisely.

## II. Related Works

### A. Weakly Supervised Object Detection

Current WSOD approaches can be mainly classified into three types: methods based on multiple instance learning (MIL), contextual information, and category activation map (CAM).

MIL-based methods were widely employed to address the WSOD task for most advanced research. WSDDN [7] was the pioneering model to apply MIL for WSOD in an end-to-end manner, successfully integrating MIL with the CNN. However, it tended to select a salient part of the object rather than the entire one as the detection result. To solve

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the problem of WSDDN, online instance classifier refinement (OICR) [8] creatively integrated multistage classifiers into WSDDN, constructing the online optimization of the instance classifier model. It led to the development of several OICR-based WSOD methods like [9], [10], which introduced more flexible and reliable instance mining strategies. For example, proposal cluster learning (PCL) [9] introduced the clustering idea on the basis of OICR [8] to solve the OICR problem of missing detection when multiple objects of the same category were present. In addition, min-entropy latent model (MELM) [10] proposed an MELM optimized with a recurrent learning algorithm. Cyclic bootstrap labeling (CBL) [28] generated pseudo-labels using image-level annotations by combining a multiple instance detection network (MIDN) with ordered online refinement. However, these approaches primarily focused on mining the most confident instance and overlooked the extraction of other instances of the same class presented in the image.

Contextual information-based methods mitigated the above MIL problems to some extent. TS$^2$C [26] creatively mined tight object frames by using the surrounding segmentation context. Kosugi et al. [29] proposed context-aware positive markers and spatially constrained negative markers. Ren et al. [30] proposed a context-centered instance-aware WSL framework.

The introduction of the CAM brought much attention to CAM-based WSOD methods. From the initial class activation mapping model proposed by Zhou et al. [31] to a series of algorithms such as Grad-CAM [32], Grad-CAM++ [33], Smooth Grad-CAM++ [34], and Score-CAM [35], divergent activation network (DANet) [36], erasing integrated learning (EIL) [37], RefCLIP [38], and so on reflected the enhancement of the detection accuracy on the CAM algorithms from different perspectives.

### B. WSOD in RSIs

With the proliferation of weakly supervised generic object detection methods, WSOD in RSIs has gained much attention and evolved into a well-studied research field in these years.

In terms of MIL, dynamic course learning (DCL) [4] dynamically assigned easy-to-difficult labels to the training samples and iteratively trained detectors under the supervision to mine high-quality instances. The multiple instance graph (MIG) [5] combined a spatial graph-based voting (SGV) mechanism and an appearance graph instance mining (AGIM) model to detect all potential object instances with the same category and thus found high-quality objects.

In terms of methods based on contextual information, progressive contextual instance refinement (PCIR) [3] focused on the entire object and implemented a progressive object box self-pruning technique to eliminate redundant candidate boxes, while the triple context-aware network (TCANet) [12] tackled the issue of grouped instances in RSIs by using a TCANet, which fully utilized the global scene and internal context information of the object instances. Coincidentally, in the fully supervised methods, the adaptive balanced network (ABNet) [1] designed an improved channel attention mechanism and an adaptive FPN, as well as a context enhancement module, effectively addressing the issues of complex backgrounds and extremely imbalanced scale distributions in remote sensing object detection. Dual-perspective alignment learning (DPAL) [2] addressed the performance limitations of existing anchor-based detectors in multimodal scenarios by designing cross-modal alignment modules and task-specific alignment modules.

In addition, the self-supervised adversarial and equivariant network (SAENet) [13] introduced an adversarial dropout-activation block along with a self-supervised transformation equivariance mechanism. Self-guided proposal generation (SGPG) [14] excavated more reliable candidate boxes, generating high-quality instance-level features explicitly. RINet [6] facilitated rotation-invariant learning of the model by detecting its predictions for prerotation and postrotation instances through multibranch online detector optimization. Mining pseudo-instance soft labels (PSLs) [17] solved part domination and misclassification due to copying and surrounding instances' pseudo-labels. The weakly supervised-oriented detector (WSODet) [16] achieved a bounding box with oriented predictions using only image-level labels. The self-training-based semantic-balanced network (STSNet) [19] and progressive image-level and instance-level feature refinement (PILFR) [20] networks solve missing detection and partial domination problems in an end-to-end manner. FGSIM [39] proposed a feature-guided instance mining and task alignment focal loss method to address the insufficiency of instance selection and consistency between classification and regression in existing models. To address WSOD's issue of partial dominance, IPS-YOLO [40] employed an improved pseudo-label generation strategy and an iterative refinement scheme, while PDEF [41] proposed an active learning strategy based on low-shot and an enhanced fine-tuning mechanism.

Different from the above methods, our approach keeps the balance between contextual information for the grouping instance problem and multiple feature extraction for the multiscale object problem in RSIs.

### C. Segment Anything Model

The recent SAM [42] has garnered significant attention within the research community. Following the model-in-the-loop manner, the SAM was trained on the SA-1B dataset containing over 1 billion masks. Moreover, SAM exhibited remarkable zero-shot transfer capabilities, leading to its integration into various visual tasks. For instance, FGVP [43] leveraged the SAM for zero-shot fine-grained visual prompting, while MedSAM [44] integrated the SAM into a large-scale medical dataset, facilitating the development of a robust medical foundation model. Additionally, several methodologies [45], [46], [47] employed the SAM to address challenges in weakly supervised semantic segmentation.

As an enhancement of state-space models (SSMs) [48], Mamba [49] showed linear complexity in input size and solved the computational efficiency problem of the Transformer on long sequences of modeling state space. Vision Mamba [50] utilizes the bidirectional SSM instead of the self-attention

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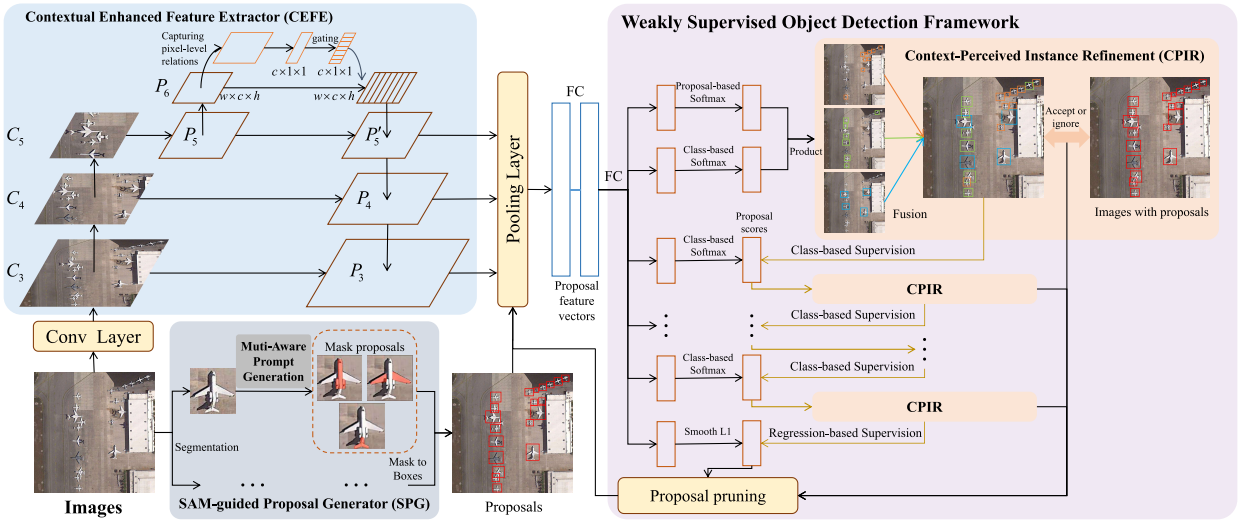



Fig. 2. Illustration of the proposed SCIR. SCIR is mainly composed of the WSOD framework with the CPIR module, CEFE, and SPG with MAPG.

mechanism for efficient visual representation learning. Inspired by the SAM and Mamba, RWKV-SAM [51] explored a high-quality and efficient SAM that combines Mamba and RWKV architectures, designing a hybrid backbone and an efficient decoder to obtain high-quality masks by utilizing multiscale markers.

In terms of SAM's applications across various remote sensing tasks, PointSAM [52] proposed a point-supervised segmentation model to improve the segmentation performance of RSIs by fine-tuning the SAM using point annotation and a self-training framework. SPFS [53] proposed a SAM-induced pseudo-fully supervised learning method for WSOD in RSIs to solve the problems of mining seed instances and positioning accuracy. CS-WSCDNet [54] achieved WSL-based change detection in RSIs by integrating the CAM and SAM, enabling pixel-level detection results without the need for pixel-level labels.

Different from the above SAM and traditional WSL, Weak-SAM [55] innovatively integrated the SAM and WSL to overcome the redundancy problem of WSOD proposals and made two key strategies to further address WSOD issues, such as pseudo-ground-truth (PGT) incompleteness and noisy PGT instances. However, it is essential to acknowledge that the SAM relies heavily on human prompts, positioning it as an interactive segmentation method. SAM-COD [56] improved the mask prediction quality of the SAM model in camouflage scenarios by introducing prompt adapters, response filters, and semantic matcher modules, as well as a new prompt adaptive knowledge distillation strategy. While both leverage SAM for weakly supervised tasks, SAM-COD focused on refining SAM's segmentation capability for camouflaged objects, whereas our SCIR repurposed SAM as an automated proposal generator for WSOD, emphasizing context-aware refinement and multiscale feature enhancement.

In our approach, we innovatively extract region proposals from the SAM, prompted automatically by classification clues. It yields high-recall proposals that outperform conventional methods such as selective search (SS) [57] in both efficiency and effectiveness.

## III. Methodology

### A. Overview of the Proposed Method

The overview of the proposed SCIR is outlined in Fig. 2. The SPG module first automatically generates prompts from spatial samples and classification clues and then forward these prompts to generate SAM proposals. Moreover, the SPG module is mainly implemented by multiple MAPG modules to generate proposals jointly. The CEFE module initially captures the global context of the visual scene to establish long-range dependencies, followed by capturing channel-wise interdependencies within this context to enhance instance-specific features effectively. Both features and proposals are finally fed into the WSOD framework. In the WSOD framework, the CPIR module extracts precise bounding boxes by integrating contextual information from surrounding regions, while identifying instances in images by assessing the contextual similarity between the top-scoring and other proposals. During the refinement process, CPIR employs regression-based self-correction to eliminate redundant proposals through a process termed "proposal pruning." Specifically, we compute the intersection-over-union (IoU) between adjacent proposals and merge those with a high score of IoU while suppressing low-confidence candidates.

### B. SAM-Guided Proposal Generator

Previous WSOD approaches, for example, SS [57] and EdgeBox [58], encounter optimization challenges due to redundant proposals, which solely rely on low-level features and limit object coverage. Formally, for object $o$ in image $I$, the proposal quality $Q$ is bounded by: $Q_{\mathrm{SS}} \leq \mathbb{E}_{p\sim\mathcal{P}_{\mathrm{SS}}}[\mathrm{IoU}(p,o)] \cdot \mathbb{I}_{\{\mathrm{scale}(p)\in[s_{\min},s_{\max}]\}}$, where $\mathrm{scale}(p)$ denotes proposal size, constrained by heuristic thresholds. To tackle this issue, the SAM utilizes mask-based instance representation: $Q_{\mathrm{SAM}} = \mathbb{E}_{m\sim\mathcal{M}_{\mathrm{SAM}}}[(\mathrm{Area}(m\cap o)/\mathrm{Area}(m\cup o))] \times$

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TABLE I
COMPARISONS OF COMPUTATIONAL COST AND RECALL (%) FOR PROPOSALS ON THE DOTA AND DIOR TESTING DATASETS

| Methods | Proposal | Inference Time (ms) | Para. (M) | FLOPs (G) | Recall of DOTA | | | | Recall of DOIR | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | IoU = 0.50 | IoU = 0.75 | IoU = 0.90 | mAP | IoU = 0.50 | IoU = 0.75 | IoU = 0.90 | mAP |
| Baseline | SS [57] | 22.4 | 138 | 147.4 | 44.12 | 27.49 | 9.15 | 26.16 | 42.03 | 26.19 | 8.72 | 24.92 |
| Ours | SS [57] | 23.6 | 145 | 142.1 | 47.03 | 29.23 | 11.75 | 27.89 | 44.95 | 28.57 | 11.04 | 26.17 |
| Ours | SAM [42] | 25.8 | 162 | 134.8 | **48.71** | **31.59** | **24.51** | **30.06** | **46.77** | **30.33** | **23.53** | **29.36** |

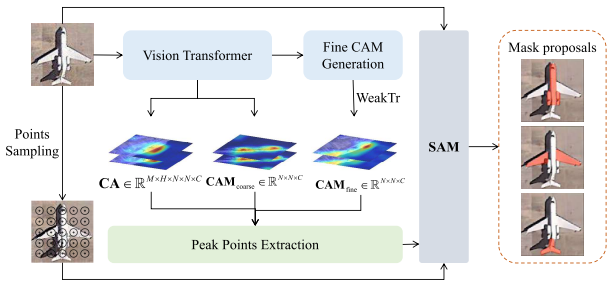



Fig. 3. Illustration of the MAPG in the SPG. It introduces classification clues and spatial points as automatic prompts for the SAM, thereby resolving the issue of the SAM's dependency on interactive prompts.

$\mathbb{P}(\text{boundary}(m) = \partial o) \geq Q_{\text{SS}} + \Delta_{\text{scale}}$, where $\Delta_{\text{scale}}$ quantifies the gain for extreme-scale objects. This theoretically ensures that SAM masks provide higher recall for objects, which can be proved in Table I.

Therefore, we propose leveraging knowledge transfer from a foundation model, namely SAM, to generate high-quality proposals, which we call SPG. To improve proposals' quality while minimizing the complexity of the SPG process as much as possible, we mainly utilize a pretrained weakly supervised semantic segmentation network WeakTr [59], to assist the SAM in generating proposals. It employs classification cues to automatically prompt the SAM, thus addressing SAM's reliance on interactive prompts.

MAPG is the most important part of our SPG. As illustrated in Fig. 3, we derive classification cues from a classification Vision Transformer (ViT). Here, the SAM leverages a pretrained ViT to generate high-quality object masks like a plane patch from the CAM. This strategy replaces manual prompting with classification-derived cues, enabling efficient, scale-invariant object patch extraction for WSOD. Specifically, we utilize the WeakTr [59] network for its superior localization capability. Initially, the MAPG extracts cross-attention maps $\mathbf{CA} \in \mathbb{R}^{M\times H\times N\times N\times C}$ from the self-attention maps, where $M$ denotes the number of transformer encoding layers, $H$ represents the number of attention heads in each layer, $N \times N$ denotes the spatial size of visual markers, and $C$ indicates the number of classification categories. Subsequently, coarse CAM, $\mathbf{CAM}_{\text{coarse}} = \text{ReLU}(\mathbf{W}_c * \mathbf{F}_{\text{last}}) \in \mathbb{R}^{N\times N\times C}$ is obtained via the convolutional CAM head, which processes visual markers at the last Transformer layer $\mathbf{F}_{\text{last}}$ with the corresponding weighting coefficient $\mathbf{W}_c$ to generate coarse CAM. Here, only less detail or precision is used in the process of visual markers. Lastly, WeakTr [59] is employed to produce fine CAM, $\mathbf{CAM}_{\text{fine}} = \text{Weaktr}(\mathbf{F}_{\text{last}}; \theta) \in \mathbb{R}^{N\times N\times C}$ by a lot of detail or precision in processing visual markers, where $\theta$ represents the learning parameters for high-detail localization.

Prompts are derived from dense sampling points, cross-attention maps, and CAMs. Dense sampling divides the image into $S \times S$ segments, using center points as prompts to ensure coverage of all regions and reduce reliance on CAM-activated areas. However, higher $S$ values may lead to invalid sampling due to the absence of direct object References. Peak points from cross-attention maps serve as instance-aware prompts, while semantic-aware prompts are extracted from both coarse and fine CAMs, focusing on foreground regions.

To address the limitations of incomplete or inaccurate CAMs, the MAPG combines spatial, semantic, and instance-aware clues, reducing reliance on a single source. This multimodal prompting strategy aids in distinguishing small, dense objects of the same class. The MAPG identifies spatial proposal regions, refines them with semantic clues, and resolves ambiguities using SAM's segmentation for robust object separation.

During the SAM-guided proposal generation phase, these 3 prompt types are used to automatically prompt the SAM. Semantic-aware and spatial-aware prompts are added directly for clear localization and foreground object focus, while redundant instance-aware prompts are filtered through clustering. Redundant peaks from cross-attention maps are filtered via distance-based clustering to merge proximate points into representative prompts, avoiding duplicate SAM-generated masks. To handle multiple duplicate SAM-generated masks, masks with significant spatial overlap are grouped using density-based clustering. For each cluster, a minimal bounding box is computed by taking the union of all mask pixels and applying convex hull algorithms, with size adaptively adjusted based on mask density. Unmergeable masks are retained as independent proposals and passed to CPIR, which refines proposals using contextual information, thereby balancing recall and precision.

Ultimately, the SAM utilizes all prompts to generate proposals. SAM-generated masks, initially part-focused, are converted to bounding boxes using a density-adaptive oriented bounding box transformation, which mitigates partial activation issues and ensures compatibility with weakly supervised pixel-level supervision, addressing remote sensing object characteristics.

### C. Contextual Enhanced Feature Extractor

To allow tiny objects to extract as much effective feature information as possible, we propose the CEFE to address the multiscale object issues in RSIs. Meanwhile, we also aim to strengthen the features of the entire object by fully utilizing the global context, ensuring that the object can be easily detected.

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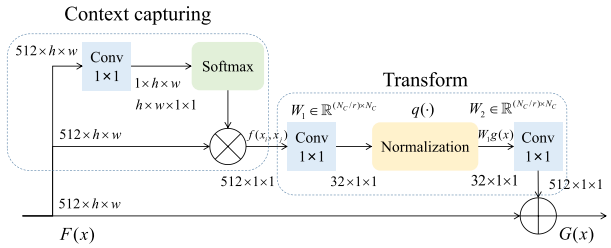



Fig. 4. Illustration of the CEFE. Adaptive acquisition of the local–global context within the visual scene enables the CEFE to emphasize regions containing instances while suppressing background regions.

Drawn on the framework of the FPN [27] to enhance pixel-level feature extraction, our approach detects objects of varying sizes at different levels of feature maps, which effectively highlights the feature information of tiny objects. Specifically, we utilize four levels of feature maps $\{P_3, P_4, P_5, P_6\}$. As depicted in Fig. 4, $P_3$, $P_4$, and $P_5$ are, respectively, derived from the feature maps $C_3$, $C_4$, and $C_5$ of backbone CNNs, followed by a $1 \times 1$ convolutional layer with top-down connections. $P_6$ is generated by applying a convolutional layer with a stride of 2 on $P_5$. As a result, the four feature levels $P_3$, $P_4$, $P_5$, and $P_6$, respectively, have strides of 8, 16, 32, and 64.

Let $F = \{x_i\}_{i=1}^{n}$ represent the feature map of $P_6$, where $n$ refers to the number of the pixels. Initially, to generate the global contextual map, we estimate and calculate the pixel-level relationship between the query and other positions in the form of

$$f(x_i, x_j) = e^{x_i^{\mathrm{T}} x_j} \tag{1}$$

where $f(\cdot)$ is a pairwise function calculating the relationship between query position $i$ and another position $j$, so that $f(x_i, x_j)$ denotes to an embedded Gaussian function. As a result, the global contextual attention map is then obtained by weighted aggregation of these pairwise relationships as follows:

$$g(x_i) = \sum_j \frac{f(x_i, x_j)}{q(x)} \cdot u(x_j) \tag{2}$$

where $q(\cdot)$ is a normalization coefficient and $u(\cdot)$ is a unary function that represents the input at position $j$. Since the fact that similar attention maps are obtained for different query positions, we can obtain the global context of any query position and share it across all positions. In contrast to traditional spatial attention models that focus only on enriching the features of local neighborhoods, we allow the global context to be captured without the additional steps as in the recursive CNN (RCNN).

Therefore, features of all positions can be enriched by taking full advantage of the above global contextual strategy. However, only using this strategy to enrich the features could inadvertently enhance background features to a certain extent. Accordingly, we aim to suppress less effective background regions by selectively emphasizing their contexts with a large amount of information and hiding them with less effect, thereby facilitating better discovery of the entire instance.

To this end, we employ incentive operations to capture channel activation by utilizing a simple gating mechanism. Specifically, two belief layers with a $1 \times 1$ filter formulate a bottleneck. Therefore, by recalibrating the global contextual features with the learned activation vectors, specific categories of features can be adaptively emphasized. The collaboration between the context capture model and the incentive operation formulates our CEFE, which is illustrated in Fig. 4 and defined as follows:

$$G(x) = F(x) + W_2 \delta(W_1 g(x)) \tag{3}$$

where $\delta(\cdot)$ is the normalization function of layers and $W_1 \in \mathbb{R}^{(N_C/r)\times N_C}$ and $W_2 \in \mathbb{R}^{(N_C/r)\times N_C}$ are, respectively, the parameters of the dimension reduction and increasing layer.

Combined FPN [27] with variations of spatial and channel attention, the CEFE integrates contextual information at different scales and levels, thereby effectively highlighting each location in a specific area of the instance. The CEFE combines established attention mechanisms like point-wise and channel attention with a novel context-aware feature extraction framework tailored for WSOD in RSIs. Unlike traditional methods focusing on local neighborhoods, it improves the detection of small, multiscale objects in cluttered or occluded scenes by incorporating both global and local context.

### D. Context-Perceived Instance Refinement

As the baseline of our network, PCIR [3] is referred to as our basic framework for WSOD because of its scalability and effectiveness. PCIR [3] focuses on the entire object and uses a progressive object proposal self-pruning mechanism to remove redundant proposals. Different from PCL [9] and PCIR [3], the regression supervision branch we propose can dynamically correct the object detection frame to prune proposals in the SPG more precisely.

Given an input image $I^{(i)}$ and its proposals $P$. In the upper part of the basic WSOD framework, let $Y = [Y_1, \ldots, Y_c, \ldots, Y_C] \in \{0, 1\}$ denote the binary image-level label to indicate the presence or absence of the specific class objects in this image. Following two fully connected (FC) layers, the feature vector of each proposal is split into two streams: the upper stream categorizes each proposal into different classification containers, while the lower one is dedicated to detection. The score of each individual proposal is generated from the elements between the results of these two streams, denoted by the special symbol $\odot$, that is,

$$x_{cr} = \mathrm{softmax}(X_c) \odot \mathrm{softmax}(X_d) \tag{4}$$

where $\mathrm{softmax}(X_c) = e^{x_{ij}^c} / \sum_{k=1}^{N_C} e^{x_{kj}^c}$, $\mathrm{softmax}(X_d) = e^{x_{ij}^d} / \sum_{r=1}^{N_R} e^{x_{ir}^d}$, $x_{ij}^c$ and $x_{ij}^d$, respectively, refer to feature vectors in the classification and detection stream for each individual proposal, $\odot$ represents the elemental corresponding multiplication by two streams, and $N_C$ and $N_R$, respectively, represent the number of image categories and proposals. Then, by summing each proposal' score $x_{cr}$, the image-level score $S_c$ for the $c$th class is calculated as follows:

$$S_c = \sum_{r=1}^{N_R} x_{cr}. \tag{5}$$

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TABLE II
ZERO-SHOT SEGMENTATION PERFORMANCE OF THE SAM ON DIFFERENT DATASETS

| Dataset | $\text{IoU}_S$ | $\text{IoU}_M$ | $\text{IoU}_L$ | mIoU |
|---|---|---|---|---|
| COCO | 65.2 | 78.9 | 85.7 | 76.6 |
| DOTA | 58.1 | 72.8 | 80.3 | 70.4 |
| DIOR | 56.7 | 71.5 | 79.6 | 69.3 |

Drawn on the multiclass cross-entropy loss function, the loss function $L(Y, S_c)$ of the two-stream network can be formulated as follows:

$$L(Y, S_c) = -\sum_{c=1}^{N_C} \{Y \log S_c + (1 - Y) \log(1 - S_c)\}. \quad (6)$$

Along with our two-stream network of the WSOD framework, several CPIR branches are introduced. Each CPIR refinement branch maps the eigenvector of the proposal to an $\{N_C + 1\}$-dimensional fractional vector $x_{cr}^k \in \mathbb{R}^{(N_C+1)\times 1}, k \in (1, \ldots, K)$, where $k$ represents the $k$th refinement branch and $\{N_C + 1\}$ represents $N_C$ different object categories and a background category. In our implementation, we employ $K = 3$ refinement stages to strike an optimal balance between detection accuracy and computational efficiency, which is determined through extensive ablation studies for different cascade levels in CPIR.

In each CPIR branch, the WSOD framework utilizes the local context information to ascertain the most representative holistic object and further leverages global context information to discover other possible instances present in the image. Denote $y_{cr}^k \in (0, \ldots, N_C)$ as the pseudo-label for the $r$th proposal in the $k$th refinement branch, and the loss function for the $k$th refinement is defined as follows:

$$L_r^k = -\frac{1}{N_R} \sum_{r=1}^{N_R} \sum_{c=1}^{N_C+1} y_{cr}^k \log x_{cr}^k. \quad (7)$$

A regression branch is placed at the bottom of the basic WSOD framework after the CPIR branches in Fig. 2. To enable end-to-end training of the weakly supervised detection model and simplify the training process, we construct a single end-to-end network by directly adding the regression branch to our WSOD framework. Meanwhile, to more accurately locate object information in the feature map, this regression branch introduces regression supervision, so as to predict the coordinates of each object frame while dynamically correcting the position information of the object frames by applying weak supervision.

Let $x_{\text{reg}}$ represent proposal's score in the regression branch, so the predicted values of the regression branch model can be expressed as $z(x_{\text{reg}}) = \sigma(x_{\text{reg}})$, where $\sigma(\cdot)$ is the regression function. Then, for the whole model training process of $N$ images, the sample true values are $y = \{y^{(1)}, y^{(2)}, \ldots, y^{(N)}\}$, and the predicted values of the object detection model are $\{z(x_{\text{reg}})\} = \{z(x_{\text{reg}}^{(1)}), z(x_{\text{reg}}^{(2)}), \ldots, z(x_{\text{reg}}^{(N)})\}$. Consequently, we derive the corresponding Smooth L1 loss function $L_{\text{reg}}$, that is,

$$L_{\text{reg}} = \frac{1}{N} \sum_{i=1}^{N} \begin{cases} 0.5[y^{(i)} - z\left(x_{\text{reg}}^{(i)}\right)]^2, & \text{if } \left|y^{(i)} - z\left(x_{\text{reg}}^{(i)}\right)\right| < 1. \\ \left|y^{(i)} - z\left(x_{\text{reg}}^{(i)}\right)\right| - 0.5, & \text{if } \left|y^{(i)} - z\left(x_{\text{reg}}^{(i)}\right)\right| \geq 1. \end{cases} \quad (8)$$

Therefore, we ultimately train the detection network by combining the loss functions of $L(Y, S_c)$, $L_r^k$, and $L_{\text{reg}}$, obtaining the total loss function as follows:

$$L = L(Y, S_c) + \sum_{k=1}^{K} L_r^k + L_{\text{reg}} \quad (9)$$

where $K$ represents the total number of refinement times.

With the optimization of the total loss, the refinement and regression branches offer extra supervision, helping avoid poor local minima; thus, the object detector can more accurately and dynamically correct the proposals during the training process, which improves the detection performance of tiny objects.

As shown in Algorithm 1, for each refinement branch, the filter process of multiple fused proposals is performed in multiple stages. First, contextual similarity $\text{sim}(p_i, p_j) = (p_i^{(f)^{\text{T}}} \cdot p_j^{(f)} / \|p_i^{(f)}\| \|p_j^{(f)}\|)$ between proposals $p_i$ and $p_j$ is computed using cosine similarity, where $p^{(f)}$ is their corresponding feature vector extracted from contextual enhanced features in the CEFE. Then, proposals with IoU above $\theta_{\text{IoU}} = 0.5$

**Algorithm 1** Proposal Filtering Algorithm in CPIR

**Input:** Set of proposals $P = \{p_1, p_2, \ldots, p_N\}$ from SPG, feature map $F$ from the CEFE, number of stages $K$
**Output:** Filtered set of proposals $P_{\text{filtered}}$

$P_{\text{current}} \leftarrow P$
**for** $k = 1$ to $K$ **do**
  $P_{\text{new}} \leftarrow \emptyset$
  **for** each proposal $p_i$ in $P_{\text{current}}$ **do**
    **if** $p_i$ is already processed **then**
      **skip**
    **end if**
    Find candidate set $U = \{p_j \in P_{\text{current}} \mid \text{IoU}(p_i, p_j) > \theta_{\text{IoU}} \text{and } \text{sim}(p_i, p_j) > 0.5\}$
    **if** $U$ is not empty **then**
      Merge all $p_j$ in $U$ with $p_i$ into a new proposal $p_{\text{merged}}$
      Update confidence of $p_{\text{merged}}$ as max one in $U$
      Add $p_{\text{merged}}$ to $P_{\text{new}}$
    **else**
      **if** confidence$(p_i) > \theta_{\text{conf}}$ **then**
        add $p_i$ to $P_{\text{new}}$
      **else**
        ignore $p_i$
      **end if**
    **end if**
  **end for**
  $P_{\text{current}} \leftarrow P_{\text{new}}$
**end for**
$P_{\text{filtered}} \leftarrow P_{\text{current}}$
**return** $P_{\text{filtered}}$

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are considered candidates for merging, while proposals with scores below $\theta_{\text{conf}} = 0.3$ are preferentially ignored. Here, cases with low IoU or similarity are handled by individual confidence checks to prevent loss of isolated objects while suppressing noise. This entire process is integrated with a regression branch for further pruning.

Before pruning, RSIs' dense object distributions amplify WSOD's proposal redundancy. To better refine the proposals, CPIR's pruning exploits spatial-contextual relationships intrinsic to RSIs, where clustered objects exhibit predictable positional correlations to distinguish true redundancies from critical instances. As a core function after several CPIR branches, the "proposal pruning" process in Fig. 2 entails an iterative and context-aware filtering mechanism designed to eliminate low-quality and redundant proposals. Rather than relying solely on proposal scores, it first identifies confident proposals and analyzes the contextual information from their surrounding regions. Then, it computes IoU between adjacent proposals, merging those with high overlap while suppressing low-confidence candidates. To avoid mistakenly merging adjacent objects, it is combined with a regression branch that provides dynamic, self-correcting supervision to refine proposal coordinates, progressively pruning partial, oversized, or duplicate boxes across multiple refinement stages to shift the detector's focus to high-quality, entire object instances. This transforms conventional class-agnostic pruning into a topology-aware optimization.

To avoid two adjacent objects being mistakenly merged into one, CPIR adopts a multistage contextual reasoning process, leveraging both local spatial relationships and global semantic consistency. Specifically, it leverages contextual dissimilarity between merged regions and their surroundings and utilizes regression-based self-correction to shrink oversized proposals. Compared with the cascade R-CNN [60], CPIR iteratively refines proposals via multistage scoring, which is guided by SAM's high-quality masks. For several adjacent tiny objects, CPIR exploits multiscale features from the CEFE to distinguish them.

## IV. Experiments

### A. Datasets and Evaluation Metrics

Our network is evaluated on the widely used DOTA [61] and DIOR [62] datasets. DOTA [61] is a well-established benchmark dataset in remote sensing, with most experiments in the field utilizing either version 1.0 or 1.5. DOTA-v1.0 includes 15 object categories and 2806 images, comprising a total of 188 282 instances. The image resolutions range from $800 \times 800$ to $4000 \times 4000$. Compared with DOTA-v1.0, DOTA-v1.5 introduces an additional category "container crane (CC)" and increases the total number of object instances to 403 318. Our experiments are primarily finished on DOTA-v1.5. DIOR [62] is a well-known public dataset for fully supervised object detection in RSIs, but it has been rarely explored in WSOD due to its significant challenges. It covers 20 different categories and contains 23 463 images (including 11 725 training images and 11 738 test images) at a resolution of $800 \times 800$ with 192 472 instances.

Both datasets are divided into three subsets: training, validation, and testing. It is worth noting that nearly every image in these two datasets contains multiple instances with varying orientations. In accordance with WSOD's standard process, our network is trained on both the training and validation (referred to as the train–val) sets, as well as evaluated on the testing set. At the same time, only image-level labels are available during the model training phase. Mean average precision (mAP) and correct localization (CorLoc) accuracy are used to assess the accuracy of object detection and localization, respectively. All metrics are performed on the PASCAL criteria with an IoU threshold of 50%.

### B. Implementation Details

In terms of initialization, VGG16 [63], ResNet-50 [64], ViT [65], and Vision Mamba [50] pretrained on the ImageNet [66] are adopted as the backbone. Experiments are conducted using an SGD optimizer on an RTX 4090 GPU, which is on average about 50% stronger than the computational efficiency, including inference time, frames per second (FPS), and FLOPs of the RTX 3090 GPU. Newly added layers are all initialized using a Gaussian distribution with a mean of 0 and a standard deviation of 0.01. To protect the features of tiny objects, we replace the pooling4 and its subsequent convolutional layers with dilated convolution layers. Before training the network, our SPG generates approximately 2000 proposals for each image. To augment the training data, we apply 3 rotation transformations {90°, 180°, 270°}.

During training, the network undergoes 300 K iterations, while the SAM and WeakTr in SPG undergo pretraining with 30 K iterations. Meanwhile, we keep the training settings including learning rate, mini-batch, weight decay, and momentum consistent with those in [8], [9], and [10]. They are set to 0.001, 2, 0.005, and 0.9, respectively. In particular, the learning rate is initially set to 0.001 and is reduced by a factor of 10 after every 100 and 10 K iterations. For each iteration, the mini-batch size is set to 2 for the stochastic gradient descent optimizer. Nonmaximum suppression (NMS) [67] is applied with a confidence threshold of 0.3 to eliminate duplicated bounding boxes.

While the SAM was only trained on natural images before, its zero-shot segmentation capabilities partially generalize to RSIs due to its ability to segment arbitrary shapes. In our experiments on the DOTA [61] and DIOR [62] datasets, SAM-generated masks for small objects or occluded instances exhibited only a little lower IoU compared to large, distinct objects.

### C. Comparison Experiments

*1) Comparison of Different Proposal Generation Methods:* To further analyze the improvements brought by the proposed MAPG in our SPG, we conduct comparison experiments for different proposal generation methods in Table I. Here, we take the SS [57] for proposals generation as the baseline method. For example, when only using our method with the same SS [57], the generated proposals can, respectively, achieve 2.60% and 2.32% higher recall (IoU = 0.90), while 1.63% and 1.25%

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TABLE III
AVERAGE PRECISION (%) AND FPS (MS) RESULTS FOR DIFFERENT BACKBONES ON THE DOTA AND DIOR TESTING DATASETS

| Methods | Backbone | FPS | DOTA | | | | DIOR | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | $AP_S$ | $AP_M$ | $AP_L$ | **mAP** | $AP_S$ | $AP_M$ | $AP_L$ | **mAP** |
| PCIR [3] | VGG16 | 6.24 | 9.75 | 30.86 | 37.87 | 26.16 | 11.61 | 28.03 | 35.12 | 24.92 |
| | ResNet-50 | 9.43 | 10.70 | 31.05 | 38.67 | 26.81 | 12.20 | 28.27 | 35.54 | 25.34 |
| | Vision Transformer | 4.94 | 10.71 | 31.74 | 38.37 | 26.94 | 12.13 | 28.06 | 35.56 | 25.25 |
| | Vision Mamba | 9.02 | 8.60 | 28.54 | 36.15 | 24.43 | 10.71 | 25.67 | 33.43 | 23.27 |
| RINet [6] | VGG16 | 2.51 | 10.99 | 34.78 | 42.67 | 29.48 | 12.16 | 32.27 | 40.31 | 28.25 |
| | ResNet-50 | 5.12 | 11.38 | 35.36 | 43.31 | 30.02 | 12.46 | 32.56 | 41.05 | 28.69 |
| | Vision Transformer | 1.80 | **11.91** | 35.67 | 43.58 | 30.39 | 12.19 | 33.13 | 40.84 | 28.72 |
| | Vision Mamba | 6.92 | 10.20 | 33.38 | 41.31 | 28.30 | 10.19 | 32.26 | 39.57 | 27.34 |
| PSL [17] | VGG16 | 2.53 | 11.10 | 35.15 | 43.12 | 29.79 | 12.93 | 32.38 | 40.52 | 28.61 |
| | ResNet-50 | 5.16 | 11.67 | 35.87 | 43.60 | 30.38 | 13.55 | 32.98 | 41.18 | 29.24 |
| | Vision Transformer | 1.85 | 11.18 | **35.97** | 43.16 | 30.10 | 13.32 | 32.91 | 41.14 | 29.12 |
| | Vision Mamba | 5.42 | 10.43 | 33.65 | 41.53 | 28.54 | 11.65 | 30.71 | 38.98 | 27.11 |
| STSNet [19] | VGG16 | 2.89 | 10.95 | 34.67 | 42.54 | 29.39 | 12.09 | 31.59 | 39.59 | 28.09 |
| | ResNet-50 | 5.89 | 11.78 | 35.65 | 42.63 | 30.02 | 12.75 | 32.02 | 40.06 | 28.27 |
| | Vision Transformer | 2.46 | 11.88 | 34.72 | 43.42 | 30.01 | 12.28 | 32.59 | 39.66 | 28.18 |
| | Vision Mamba | 7.70 | 10.32 | 33.35 | 41.45 | 28.37 | 11.11 | 30.11 | 37.91 | 26.38 |
| Ours | VGG16 | 2.02 | 11.20 | 35.46 | 43.51 | 30.06 | 13.45 | 32.96 | 41.67 | 29.36 |
| | ResNet-50 | 4.12 | 11.83 | 35.57 | 44.35 | **30.58** | **14.37** | **33.95** | **42.95** | **30.42** |
| | Vision Transformer | 1.73 | 11.22 | 35.86 | **44.38** | 30.49 | 13.80 | 33.88 | 42.25 | 29.98 |
| | Vision Mamba | 4.95 | 10.30 | 34.01 | 42.12 | 28.81 | 12.55 | 30.99 | 39.95 | 27.83 |

higher mAP for the DOTA [61] and DIOR [62] datasets. When only changing the proposal generation method from SS [57] to the SAM in our SCIR framework, the generated proposals can, respectively, achieve 12.76% and 12.49% higher recall (IoU = 0.90), while 2.17% and 3.19% higher mAP for the DOTA [61] and DIOR [62] datasets. It is obvious that compared to other existing large model-based methods in the WSOD field, the pretrained weakly supervised semantic segmentation model and the SAM in the SPG module yield high-recall proposals to facilitate subsequent model training and enhance its object detection performance much more with only a little complexity of modules and processing. In terms of computational overhead and complexity, SAM integration increases inference time by 15.2% from 22.4 to 25.8 ms compared to the baseline, but reduces FLOPs by 8.6% from 147.4G to 134.8G, which is due to efficient mask processing. As the SAM is added, the computational parameter of the model slightly increases by about 24 M, but its dramatic recall gains and reduced FLOPs make it a net positive for WSOD. Compared with the traditional SS [57] and SAM [42], SPG's clustering reduces SAM prompts, which can scale better to larger datasets or real-time applications. The trade-off favors accuracy-critical applications, with optimizations such as pruning and hardware easing deployment constraints, which is acceptable considering the significant performance improvements.

By clustering and fusing SAM prompts, the SPG ensures that even if CAMs partially fail, such as missing small objects or activating backgrounds, the other prompt streams compensate. Shown in Table I, this redundancy can be validated where the SAM achieves 24.51% recall at IoU = 0.9 on DOTA [61], far surpassing 11.75% of SS [57], indicating robustness to partial CAM failures. Compared to other existing large model-based methods in the WSOD field, the pretrained weakly supervised semantic segmentation model and SAM in the SPG module yield high-recall proposals to facilitate subsequent model training and improve its object detection performance much more with only increasing little complexity of modules and processing.

*2) Comparison of SAM's Zero-Shot Segmentation Capabilities:* To validate the zero-shot transfer capability of the SAM on RSIs, we conduct comprehensive evaluations on the DOTA and DIOR testing datasets. We randomly select 500 object instances from each dataset, covering all categories and scale variations. Each instance is cropped with its ground-truth bounding box and processed by the SAM using automatically generated point prompts at the object center.

The segmentation performance is measured by IoU between SAM-generated masks and ground-truth masks. For comparison, we also evaluate the SAM on the COCO dataset under the same setting. Table II shows the zero-shot segmentation performance of the SAM on three different datasets. Here, $IoU_S$, $IoU_M$, and $IoU_L$, respectively, denote the IoU of small, medium, and large objects. As indicated in Table II, the SAM achieves slightly lower IoU on RSIs compared to natural images, with a more significant drop for small objects (up to 9.5% absolute IoU decrease on DIOR [62]). This performance degradation aligns with our observation that SAM's mask decoder tends to under-segment small or densely clustered objects in RSIs due to their distinct textures and complex backgrounds.

*3) Comparison of Different Backbones:* Considering that the WSOD methods with only VGG16 [63] backbone may further limit the performance of WSOD in RSIs, we perform comparative experiments with ResNet-50 [64], ViT [65], and Vision Mamba [50] on both the DOTA [61] and DIOR [62] datasets. To evaluate the computational speed of the object detection model, we adopt the index of FPS, which represents the average number of image frames processed per second.

As shown in Table III, compared with VGG16 [63], the computational speed FPS of ResNet-50 [64], ViT [65], and

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TABLE IV

AVERAGE PRECISION (%) RESULTS FOR DIFFERENT METHODS ON THE DOTA TESTING DATASET. (THE BEST AND SECOND-BEST PERFORMANCES ARE, RESPECTIVELY, HIGHLIGHTED IN RED AND BLUE)

| WSOD Methods | BD | BC | BR | GTF | HA | HC | LV | PL | RA | SH | SV | SBF | ST | SP | TC | CC | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| WSDDN (2016) [7] | 28.19 | 26.79 | 4.11 | 19.34 | 11.02 | 2.83 | 21.28 | 30.44 | 11.59 | 1.19 | 6.12 | 10.53 | 8.31 | 5.53 | 31.14 | 2.55 | **13.81** |
| OICR (2017) [8] | 33.84 | 29.61 | 5.11 | 24.07 | 13.71 | 3.52 | 26.49 | 37.88 | 14.42 | 6.46 | 7.62 | 13.10 | 10.34 | 6.89 | 38.65 | 3.18 | **17.18** |
| PCL (2018) [9] | 37.30 | 32.63 | 5.64 | 26.53 | 15.11 | 3.88 | 29.20 | 41.75 | 15.90 | 7.12 | 8.40 | 14.44 | 11.39 | 7.59 | 42.71 | 3.50 | **18.94** |
| FCC-Net (2020) [68] | 37.62 | 32.92 | 5.69 | 26.77 | 15.24 | 3.91 | 29.45 | 42.12 | 16.04 | 7.18 | 8.48 | 14.57 | 11.49 | 7.66 | 43.09 | 3.53 | **19.11** |
| PCIR (2020) [3] | 51.10 | 44.71 | 7.72 | 36.35 | 20.70 | 5.31 | 40.00 | 57.20 | 21.78 | 9.75 | 11.51 | 19.78 | 15.61 | 10.40 | 58.52 | 4.80 | **25.95** |
| DCL (2021) [4] | 41.44 | 36.26 | 6.26 | 29.48 | 16.79 | 4.31 | 32.44 | 46.39 | 17.66 | 7.91 | 9.33 | 16.04 | 12.66 | 8.43 | 47.35 | 3.89 | **21.04** |
| TCANet (2021) [12] | 52.96 | 46.33 | 8.00 | 37.67 | 21.46 | 5.51 | 41.46 | 59.28 | 22.57 | 10.11 | 11.93 | 20.50 | 16.18 | 10.78 | 60.54 | 4.97 | **26.89** |
| SAENet (2022) [13] | 55.59 | 48.64 | 8.40 | 39.55 | 22.52 | 5.78 | 43.52 | 62.23 | 23.69 | 10.61 | 12.52 | 21.52 | 16.98 | 11.31 | 63.46 | 5.22 | **28.22** |
| SGPG (2022) [14] | 54.54 | 47.65 | 8.05 | 35.11 | 19.40 | 4.93 | 42.69 | 58.64 | 22.13 | 9.47 | 10.68 | 18.79 | 15.74 | 9.75 | 55.27 | 4.33 | **26.07** |
| RINet (2022) [6] | 57.93 | 50.69 | 8.76 | 41.22 | 23.47 | 6.02 | 45.35 | 64.86 | 24.69 | 11.06 | 13.05 | 26.43 | 17.70 | 12.79 | 62.25 | 5.44 | **29.48** |
| CLN (2023) [15] | 57.52 | 32.83 | 5.67 | 36.69 | 15.20 | 3.90 | 29.37 | 42.00 | 15.99 | 7.16 | 8.45 | 14.53 | 11.46 | 7.64 | 42.87 | 3.52 | **20.92** |
| WSONet (2023) [16] | 55.93 | 53.93 | 4.45 | 42.79 | 22.66 | 5.82 | 43.78 | 62.61 | 23.84 | 4.68 | 12.60 | 21.65 | 12.09 | 11.38 | 73.45 | 3.25 | **28.43** |
| PSL (2023) [17] | 61.68 | 51.34 | 8.87 | 41.74 | 23.77 | 6.10 | 45.93 | 65.68 | 25.01 | 11.20 | 13.22 | 22.72 | 17.93 | 11.94 | 63.99 | 5.51 | **29.79** |
| HIMDLA (2024) [18] | 54.35 | 48.81 | 6.70 | 41.55 | 17.97 | 4.61 | 44.72 | 59.65 | 18.90 | 18.47 | 9.99 | 17.17 | 20.55 | 9.03 | 60.79 | 4.17 | **27.34** |
| STSNet (2024) [19] | 57.61 | 50.41 | 8.71 | 40.99 | 25.34 | 5.99 | 45.10 | 64.50 | 24.56 | 11.00 | 12.98 | 22.31 | 17.60 | 11.72 | 65.98 | 5.41 | **29.39** |
| PILFR (2024) [20] | 58.24 | 49.22 | 7.88 | 41.23 | 22.22 | 5.59 | 43.37 | 61.48 | 22.78 | 11.67 | 12.12 | 21.77 | 16.99 | 11.17 | 63.82 | 4.80 | **28.40** |
| FGSIM (2025) [39] | 56.92 | 49.81 | 9.56 | 40.50 | 23.06 | 5.92 | 44.56 | 63.72 | 23.30 | 10.87 | 12.82 | 22.04 | 17.39 | 11.58 | 65.19 | 5.35 | **28.91** |
| IPS-YOLO (2025) [40] | 58.38 | 51.08 | 9.81 | 41.53 | 23.65 | 6.07 | 45.70 | 65.36 | 23.90 | 11.14 | 13.15 | 22.60 | 17.84 | 11.88 | 66.86 | 5.48 | **29.65** |
| Ours | 59.19 | 51.79 | 9.95 | 42.11 | 23.98 | 6.16 | 46.33 | 66.26 | 24.23 | 11.30 | 13.33 | 22.92 | 18.08 | 12.04 | 67.78 | 5.56 | **30.06** |

Vision Mamba [50] backbones, respectively, increases by an average of 1.9 times, decreases by 23%, and increases by 2.25 times among the four advanced methods (i.e., PCIR [3], RINet [6], PSL [17], and STSNet [19]). Take the Vision Mamba [50] compared with VGG16 [63] as an example, it increases FPS by 1.45, 2.75, 2.14, and 2.66 times for the four respective methods [3], [6], [17], [19], while the mAP, respectively, decreases by about 1.69%, 1.05%, 1.38%, and 1.37% for the two datasets. Furthermore, take VGG16 [63] as the baseline for analyzing the average improvement in mAP of the two datasets, ResNet-50 [64], respectively, improves by 0.53%, 0.49%, 0.61%, and 0.41% for the four respective methods [3], [6], [17], [19], while ViT [65] improves by 0.56%, 0.69%, 0.41%, and 0.35%. As for our SCIR, only changing the backbone from VGG16 [63] to Vision Mamba [50] can improve the computational efficiency from 2.02 to 4.95 frames/s (about 2.45 times), while the mAP, respectively, decreases 0.41% and 0.83%, and the $AP_S$ is only reduced by 0.40% for the two datasets. Compared with the above four advanced methods [3], [6], [17], [19], comparative experiments for different backbones prove that our SCIR can adapt to different types of backbones and achieve a better tradeoff between speed and accuracy, which demonstrates its generalizability.

*4) Comparison With SOTA WSOD Methods:* Compared with 18 existing methods related to WSOD in RSIs these years, we evaluate the proposed SCIR on the DOTA [61] and DIOR [62] datasets to provide comprehensive comparisons with the state-of-the-art (SOTA). Considering that some SOTA methods like RINet [6] have been tested only on the DIOR [62] dataset but not on the DOTA [61] dataset, we keep the same training settings as [8], [9], [10] mentioned above to test them on the DOTA [61] dataset.

Tables IV and V show quantitative comparisons of the mAP for each category with existing advanced methods on the DOTA [61] and DIOR [62] datasets, respectively, which shows that our SCIR, respectively, achieves the mAP of 30.06% and 29.36%. Our network outperforms SOTA methods in about a quarter of the categories, while the remaining nearly half of the categories reach the second best. Here, the best and second-best performances are, respectively, highlighted in red and blue. Specifically, thanks to the high-quality proposals generated by the SAM [42], our SCIR outperforms these advanced methods in the detection of tiny objects overall. For example, on the categories of bridge (BR) and airplane (PL), DOTA [61], respectively, achieves the mAP of 9.95% and 66.26%, which is better than that of FGSIM [39] and IPS-YOLO [40] in 2025, while DIOR [62], respectively, reaches the mAP of 10.62% and 33.06%. Table VI shows quantitative comparisons of CorLoc with existing advanced methods on the two datasets, which show that our SCIR, respectively, achieves the CorLoc of 55.77% and 53.42%. Table VII shows quantitative comparisons of parameters and FLOPs with existing advanced methods on the two datasets, which shows that our SCIR achieves 134.8G FLOPs with 162 M parameters. SCIR's 162 M parameters enable robust feature representation but exceed lightweight models like CLN [15], which positions SCIR within the mid-to-high complexity spectrum among WSOD advanced methods.

To rigorously validate SCIR's improvements, we conducted a statistical significance testing experiment for performance improvements on the two remote sensing datasets based on Tables IV and V. Statistical significance testing using per-class AP differences revealed that SCIR's improvements are scale and density-dependent. As presented in Table VIII, SCIR's performance gains are statistically significant ($p < 0.01$) across both datasets and some SOTA WSOD methods [3], [6], [16], [17], [19]. Here, all $p$-values $< 0.01$ confirm improvements are statistically significant, not due to random variation. This confirms SCIR's robustness in dense, multi-scale scenarios while highlighting class-specific optimization opportunities.

According to the average precision, CorLoc, parameters, FLOPs, and statistical significance results, comprehensive comparison experiments show that the proposed SCIR significantly surpasses existing SOTA methods and demonstrates its robustness.

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TABLE V

AVERAGE PRECISION (%) RESULTS FOR DIFFERENT METHODS ON THE DIOR TESTING DATASET. (THE BEST AND SECOND-BEST PERFORMANCES ARE, RESPECTIVELY, HIGHLIGHTED IN RED AND BLUE)

| WSOD Methods | PL | AP | BF | BC | BR | CM | DA | ESA | ETS | GC | GTF | HA | OP | SH | SD | ST | TC | TS | VH | WM | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| WSDDN (2016) [7] | 9.06 | 39.68 | 37.81 | 20.16 | 0.25 | 12.18 | 0.57 | 0.65 | 11.88 | 4.90 | 42.35 | 4.66 | 1.06 | 0.70 | 63.03 | 3.95 | 6.06 | 0.51 | 4.55 | 1.14 | **13.26** |
| OICR (2017) [8] | 8.70 | 28.26 | 44.05 | 18.22 | 1.30 | 20.15 | 0.09 | 0.65 | 29.89 | 13.80 | 57.39 | 10.66 | 11.06 | 9.09 | 59.29 | 7.10 | 0.68 | 0.14 | 9.09 | 0.41 | **16.50** |
| PCL (2018) [9] | 21.52 | 35.19 | 59.80 | 23.49 | 2.95 | 43.71 | 0.12 | 0.90 | 1.49 | 2.88 | 56.36 | 16.76 | 11.05 | 9.09 | 57.62 | 9.09 | 2.47 | 0.12 | 4.55 | 4.55 | **18.19** |
| FCC-Net (2020) [68] | 20.10 | 38.80 | 52.00 | 23.40 | 1.80 | 22.30 | 0.20 | 0.60 | 28.70 | 14.10 | 56.00 | 11.10 | 10.90 | 10.00 | 57.50 | 9.10 | 3.60 | 0.10 | 5.90 | 0.70 | **18.35** |
| PCIR (2020) [3] | 30.37 | 36.06 | 54.22 | 26.60 | 9.09 | 58.59 | 0.22 | 9.65 | 36.18 | 32.59 | 58.51 | 8.60 | 21.63 | 12.09 | 64.28 | 9.09 | 13.62 | 0.30 | 9.09 | 7.52 | **24.92** |
| DCL (2021) [4] | 20.90 | 22.70 | 54.20 | 11.50 | 6.00 | 61.00 | 0.10 | 1.10 | 31.00 | 30.90 | 56.50 | 5.10 | 2.70 | 9.09 | 63.70 | 9.10 | 10.40 | 0.00 | 7.30 | 0.80 | **20.21** |
| TCANet (2021) [12] | 25.13 | 30.84 | 62.92 | 40.00 | 4.13 | 67.78 | 8.07 | 23.80 | 29.89 | 22.34 | 53.85 | 24.84 | 11.06 | 9.09 | 46.40 | 13.74 | 30.98 | 1.47 | 9.09 | 1.00 | **25.82** |
| SAENet (2022) [13] | 20.60 | 62.40 | 62.70 | 23.50 | 7.60 | 64.60 | 0.20 | 34.50 | 30.60 | 55.40 | 52.70 | 17.60 | 6.90 | 9.09 | 51.60 | 15.40 | 1.70 | 14.40 | 1.40 | 9.20 | **27.10** |
| SGPG (2022) [14] | 31.30 | 36.70 | 62.80 | 29.10 | 6.10 | 62.70 | 0.30 | 15.00 | 30.10 | 35.00 | 48.00 | 27.10 | 12.00 | 10.00 | 60.00 | 15.10 | 21.00 | 9.90 | 3.20 | 0.10 | **25.78** |
| RINet (2022) [6] | 26.20 | 57.40 | 62.70 | 25.10 | 9.90 | 69.20 | 1.40 | 13.30 | 36.20 | 51.40 | 53.90 | 28.60 | 4.80 | 9.09 | 52.70 | 15.80 | 20.60 | 12.90 | 9.09 | 4.70 | **28.25** |
| CLN (2023) [15] | 10.10 | 33.20 | 43.90 | 23.40 | 0.80 | 38.80 | 0.70 | 1.10 | 19.30 | 11.60 | 48.90 | 19.60 | 9.50 | 13.00 | 54.50 | 10.80 | 10.30 | 0.50 | 9.20 | 6.70 | **18.29** |
| WSONet (2023) [16] | 32.20 | 53.30 | 66.50 | 76.70 | 0.10 | 57.10 | 0.10 | 0.10 | 0.40 | 42.80 | 66.60 | 0.10 | 1.90 | 2.00 | 52.60 | 22.40 | 68.80 | 0.20 | 1.20 | 0.30 | **27.27** |
| PSL (2023) [17] | 29.10 | 49.80 | 70.90 | 41.40 | 7.20 | 45.50 | 0.20 | 35.40 | 36.80 | 60.80 | 48.50 | 14.00 | 25.10 | 18.50 | 48.90 | 11.70 | 11.90 | 3.50 | 11.30 | 1.70 | **28.61** |
| HIMDLA (2024) [18] | 10.50 | 32.40 | 64.20 | 28.00 | 1.10 | 13.30 | 0.30 | 0.30 | 29.90 | 50.90 | 55.40 | 12.40 | 15.00 | 33.95 | 33.90 | 29.98 | 1.30 | 4.10 | 14.80 | 0.80 | **21.20** |
| STSNet (2024) [19] | 31.00 | 67.00 | 63.10 | 37.80 | 0.30 | 71.10 | 0.20 | 0.09 | 0.08 | 62.30 | 53.20 | 40.50 | 0.60 | 9.09 | 53.30 | 16.60 | 32.20 | 13.80 | 9.09 | 0.50 | **28.09** |
| PILFR (2024) [20] | 32.90 | 70.50 | 63.20 | 45.70 | 0.20 | 69.70 | 0.20 | 12.40 | 39.40 | 56.40 | 55.30 | 16.60 | 0.60 | 9.10 | 54.80 | 18.10 | 11.00 | 16.10 | 9.10 | 1.10 | **29.12** |
| FGSIM (2025) [39] | 37.50 | 62.70 | 60.90 | 30.20 | 9.20 | 63.40 | 1.90 | 16.80 | 37.60 | 54.70 | 57.90 | 23.70 | 24.90 | 17.20 | 55.90 | 11.30 | 12.10 | 14.90 | 12.80 | 3.60 | **29.26** |
| IPS-YOLO (2025) [40] | 55.40 | 50.24 | 71.20 | 30.50 | 1.01 | 68.20 | 1.22 | 16.05 | 37.30 | 52.68 | 61.50 | 27.30 | 1.03 | 15.20 | 54.27 | 25.50 | 24.90 | 1.30 | 12.60 | 1.25 | **29.23** |
| Ours | 33.06 | 39.49 | 61.86 | 43.10 | 10.62 | 64.92 | 1.91 | 12.37 | 44.40 | 35.49 | 62.41 | 19.23 | 23.48 | 13.76 | 69.40 | 17.58 | 15.19 | 1.43 | 9.69 | 7.81 | **29.36** |

TABLE VI

CORLOC (%) RESULTS FOR DIFFERENT METHODS ON THE DOTA AND DIOR TRAIN–VAL DATASETS

| WSOD Methods | WSDDN [7] | OICR [8] | PCL [9] | FCC-Net [68] | PCIR [3] | DCL [4] | TCANet [12] | SAENet [13] | SGPG [14] | Ours |
|---|---|---|---|---|---|---|---|---|---|---|
| DOTA | 33.79 | 36.22 | 40.25 | 37.77 | 44.87 | 41.72 | 45.83 | 47.60 | 45.91 | **55.77** |
| DIOR | 32.44 | 34.77 | 41.52 | 41.70 | 46.12 | 42.22 | 48.41 | 50.80 | 48.25 | **53.42** |
| **WSOD Methods** | RINet [6] | CLN [15] | WSODet [16] | PSL [17] | HIMDLA [18] | STSNet [19] | PILFR [20] | FGSIM [39] | IPS-YOLO [40] | Ours |
| DOTA | 55.01 | 40.27 | 53.34 | 55.43 | 49.44 | 54.78 | 55.02 | 53.64 | 55.01 | **55.77** |
| DIOR | 52.81 | 41.56 | 51.19 | 53.20 | 44.30 | 53.11 | 53.24 | 53.29 | 53.18 | **53.42** |

*5) Comparison With SAM-Guided Methods:* As for recent SAM-guided object detection methods, we select SAM-COD (2024) [56], SPFS (2024) [53], and PointSAM (2025) [52] to conduct the comparative experiment and keep the same training settings as [8], [9], [10] mentioned above to test them on the DOTA [61] and DIOR [62] datasets. In terms of technical differences compared to our SCIR, SAM-COD [56] enhances SAM's segmentation capability specifically for camouflaged objects by integrating components like prompt adapters and semantic matchers, alongside a knowledge distillation strategy, but it remains dependent on manual prompts and domain-specific tuning. SPFS [53] employs the SAM within a pseudo-fully supervised learning framework for WSOD in RSIs, aiming to address seed instance mining and positioning, though its prompt generation still relies on heuristic rules. PointSAM [52] adapts the SAM for RSI segmentation using point annotations and a self-training framework with prototype-based regularization, effectively improving performance in dense scenes but requiring point-level supervision that differs from image-level weak supervision.

Table IX shows quantitative comparisons of mAP and CorLoc results with these methods on the two datasets. Take SAM-COD [56] as an example, our SCIR's mAP, respectively, outperforms that of SAM-COD [56] by 0.86% and 0.84% on the DOTA and DIOR testing datasets. SAM-COD [56] specifically refines SAM's segmentation capabilities for camouflaged objects by addressing its limitations in extreme responses, semantic misinterpretations, and unstable feature representations. Conversely, our SCIR repurposes the SAM as an automated proposal generator for WSOD, prioritizing context-aware refinement and multiscale feature enhancement, and advances SAM-guided detection in RSIs through eliminating manual prompts via WeakTr-driven CAMs. Therefore, SCIR advances SAM-guided detection by embedding the SAM into a WSOD-specific pipeline, balancing automation, context-awareness, and scalability.

*6) Comparison on Different Domains:* To strengthen SCIR's generalizability across different dataset domains, we conduct a simple evaluation on three general object detection datasets (VOC07, VOC12, and MS COCO). As presented in Table X, our SCIR achieves 49.75% mAP on VOC07 and 46.52% on VOC12, outperforming recent WSOD benchmarks, including STSNet, by 0.65% and 0.67%, respectively. On the more challenging MS COCO, we attain 22.86% mAP, surpassing SOTA PSL by 0.32%. Compared with the advanced WSOD methods, comparative experiments for datasets across different domains prove that our SCIR transfers effectively from RSIs to natural images, outperforming domain-specific WSOD methods, which demonstrates consistent superiority beyond remote sensing domains.

## D. Ablation Studies

*1) Ablation for Different Components of SCIR:* In this section, we perform ablation experiments on both the DOTA [61] and DIOR [62] datasets to reveal the contributions of the

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TABLE VII

PARAMETERS (M) AND FLOPS (G) RESULTS FOR DIFFERENT METHODS ON THE DOTA AND DIOR DATASETS

| **WSOD Methods** | WSDDN [7] | OICR [8] | PCL [9] | FCC-Net [68] | PCIR [3] | DCL [4] | TCANet [12] | SAENet [13] | SGPG [14] | Ours |
|---|---|---|---|---|---|---|---|---|---|---|
| Parameters (M) | 121 | 126 | 131 | 134 | 138 | 140 | 142 | 145 | 148 | 162 |
| FLOPs (G) | 128.6 | 132.9 | 136.5 | 141.2 | 147.4 | 143.8 | 145.3 | 142.1 | 146.7 | 134.8 |
| **WSOD Methods** | RINet [6] | CLN [15] | WSODet [16] | PSL [17] | HIMDLA [18] | STSNet [19] | PILFR [20] | FGSIM [39] | IPS-YOLO [40] | Ours |
| Parameters (M) | 152 | 89 | 112 | 158 | 93 | 163 | 96 | 105 | 117 | 162 |
| FLOPs (G) | 149.6 | 98.2 | 121.4 | 152.3 | 105.7 | 157.2 | 110.4 | 118.9 | 125.1 | 134.8 |

TABLE VIII

STATISTICAL SIGNIFICANCE TESTING EXPERIMENT FOR PERFORMANCE IMPROVEMENTS ON THE DOTA AND DIOR DATASETS

| **WSOD Methods** | $t$-statistic | $p$-value | Significance |
|---|---|---|---|
| PCIR (2020) [3] | 12.07 | $4.3\times10^{-16}$ | $p<0.001$ |
| RINet (2022) [6] | 4.83 | $6.5\times10^{-6}$ | $p<0.001$ |
| WSONet (2023) [16] | 6.54 | $3.1\times10^{-8}$ | $p<0.001$ |
| PSL (2023) [17] | 3.41 | $1.2\times10^{-3}$ | $p<0.01$ |
| STSNet (2024) [19] | 4.12 | $1.4\times10^{-4}$ | $p<0.001$ |

TABLE IX

AVERAGE PRECISION AND CORLOC (%) RESULTS FOR DIFFERENT SAM-GUIDED METHODS ON THE DOTA AND DIOR DATASETS

| **SAM-guided Methods** | **DOTA** | | **DIOR** | |
|---|---|---|---|---|
| | **mAP** | **CorLoc** | **mAP** | **CorLoc** |
| SAM-COD (2024) [56] | 29.20 | 55.54 | 28.52 | 53.20 |
| SPFS (2024) [53] | 29.79 | 55.23 | 29.10 | 52.90 |
| PointSAM (2025) [52] | 29.71 | 55.12 | 29.02 | 52.75 |
| Ours | **30.06** | **55.77** | **29.36** | **53.42** |

TABLE X

AVERAGE PRECISION (%) RESULTS FOR DIFFERENT METHODS ON THE VOC07, VOC12, AND MS COCO TESTING DATASETS

| **WSOD Methods** | **VOC07** | **VOC12** | **COCO** |
|---|---|---|---|
| WSDDN (2016) [7] | 34.80 | 35.30 | 12.10 |
| OICR (2017) [8] | 41.20 | 37.90 | 14.30 |
| PCL (2018) [9] | 43.50 | 44.20 | 19.40 |
| PCIR (2020) [3] | 46.91 | 43.65 | 20.01 |
| RINet (2022) [6] | 49.19 | 45.94 | 22.25 |
| PSL (2023) [17] | 49.45 | 46.08 | 22.54 |
| STSNet (2024) [19] | 49.10 | 45.85 | 22.16 |
| Ours | **49.75** | **46.52** | **22.86** |

proposed SPG, CEFE, and CPIR. To validate the effectiveness of our SCIR, we adopt a training strategy similar to PCIR [3], which selects only the highest-confidence proposal to refine the subsequent classifier.

As presented in Table XI, taking the DIOR [62] dataset as an example, the integration of the SPG, CEFE, and CPIR into our SCIR alone results in respective increases in the mAP of 1.29%, 0.94%, and 1.23%, along with improvements in CorLoc of 2.41%, 1.75%, and 2.29%, which indicates that all the modules have positive effects on performance enhancement. Moreover, when selecting any two of these modules for integration into our network, the mAP, respectively, increases by 2.41%, 2.56%, and 2.27%, and CorLoc, respectively, improves by 4.47%, 4.75%, and 4.21%, which indicates that the improved accuracy of the mAP and CorLoc is basically higher than the sum of the corresponding individual modules. As for the DOTA [61] dataset, the integration of the SPG, CEFE, and CPIR into our SCIR alone yields respective increases in the mAP of 1.35%, 0.99%, and 1.29%, along with improvements in CorLoc of 2.52%, 1.83%, and 2.40%, while integrating any two of these modules into our network results in respective increases in the mAP of 2.52%, 2.68%, and 2.38%, and respective improvements in CorLoc of 4.68%, 4.98%, and 4.41%. It is obvious that the SPG, CEFE, and CPIR modules exhibit similar performance effects on the DOTA [61] dataset as previously analyzed.

The FLOPs decrease slightly with each added component, suggesting that the proposed SPG, CEFE, and CPIR lead to better computational efficiency at the expense of only a small increase in inference time. In this context, the tradeoff between slightly higher inference time and improved performance is acceptable, particularly in WSOD.

For multiscale and densely arranged object issues in RSIs, our CEFE module improves the mAP while improving the average precision of small objects ($AP_S$) compared to all SOTA methods. Here, $AP_S$, $AP_M$, and $AP_L$, respectively, denote small, medium, and large objects (they are divided by $32 \times 32$ and $96 \times 96$ pixels). Take the ablation experiment of the CEFE module presented in Table XI as an example, where it, respectively, reaches $AP_S$ exceeding 11% and 13% on the DOTA [61] and DIOR [62] datasets, demonstrating that the CEFE effectively enhances the performance of small-object detection under dense arrangement. For more densely arranged objects, SCIR outperforms the baseline by 1% and 1.2% $AP_S$ by resolving part ambiguities via SPG prompts and CPIR's local-context suppression. Comprehensive ablation experiments demonstrate that the three proposed modules of our SCIR enhance the WSOD's effectiveness in RSIs.

*2) Parameter Analysis:* In this section, we conduct the ablation experiments to analyze the impact of $S$ and Top-K in the SPG on the performance of the proposed network. The hyperparameter $S$ controls the spatial grid granularity for prompt generation in the SPG, directly affecting proposal quantity and quality. Top-K is the specific number of peak points in the SAM prompts extraction phase.

As shown in Fig. 5(a), our method achieves a peak 72.8% Mask IoU and 38.7 patches per RSI with $S = 8$, maximizing both proposal quality and final detection accuracy (i.e., 30.06% mAP on DOTA [61] and 29.36% mAP on DIOR [62]). This configuration generates sufficient prompts to cover objects across scales while avoiding redundancy. Specifically, as the

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TABLE XI

ABLATION RESULTS (%) FOR DIFFERENT COMPONENTS OF SCIR ON THE DOTA AND DIOR DATASETS

| Modules | | | Inference | FLOPs | DOTA | | | | | DIOR | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SPG | CEFE | CPIR | Time (ms) | (G) | $AP_S$ | $AP_M$ | $AP_L$ | mAP | CorLoc | $AP_S$ | $AP_M$ | $AP_L$ | mAP | CorLoc |
| Baseline PCIR [3] | | | 22.4 | 147.4 | 9.75 | 30.86 | 37.87 | 26.16 | 48.41 | 11.61 | 28.03 | 35.12 | 24.92 | 46.12 |
| ✓ | | | 25.1 | 143.5 | 10.25 | 32.46 | 39.82 | 27.51 | 50.93 | 12.21 | 29.48 | 36.94 | 26.21 | 48.53 |
| | ✓ | | 25.2 | 143.1 | 11.12 | 31.53 | 38.80 | 27.15 | 50.24 | 13.05 | 28.58 | 35.94 | 25.86 | 47.87 |
| | | ✓ | 24.7 | 143.9 | 10.23 | 32.39 | 39.73 | 27.45 | 50.81 | 12.19 | 29.41 | 36.85 | 26.15 | 48.41 |
| ✓ | ✓ | | 25.6 | 137.7 | 11.19 | 33.59 | 41.26 | 28.68 | 53.09 | 13.24 | 30.49 | 38.27 | 27.33 | 50.59 |
| ✓ | | ✓ | 25.3 | 138.5 | 10.75 | 34.03 | 41.75 | 28.84 | 53.39 | 12.81 | 30.91 | 38.73 | 27.48 | 50.87 |
| | ✓ | ✓ | 24.9 | 139.1 | 11.14 | 33.42 | 41.06 | 28.54 | 52.82 | 13.17 | 30.33 | 38.07 | 27.19 | 50.33 |
| ✓ | ✓ | ✓ | 25.8 | 134.8 | **11.20** | **35.46** | **43.51** | **30.06** | **55.77** | **13.45** | **32.96** | **41.67** | **29.36** | **53.42** |

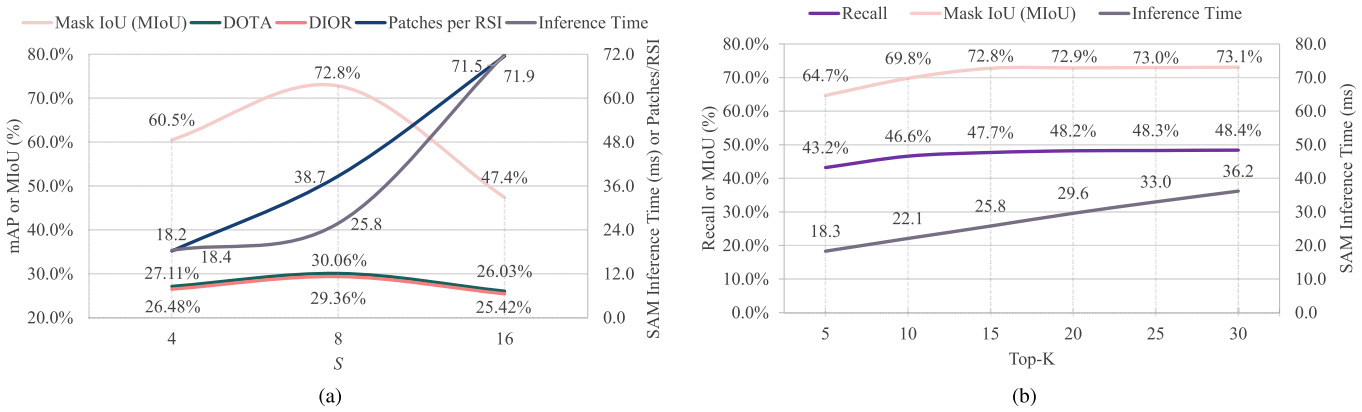



Fig. 5. Ablation results for parameter $S$ and Top-K in the SPG on the DOTA and DIOR datasets. (a) Impact of $S$ values on the number and quality of generated object patches per RSI (Top-K = 15). (b) Impact of Top-K values on the proposal quality in the Top-K peak points extraction process ($S$ = 8).

TABLE XII

ABLATION RESULTS (%) FOR DIFFERENT CASCADE LEVELS $K$ [SEE (9)] IN CPIR'S REFINEMENT BRANCHES ON THE DOTA AND DIOR DATASETS

| $K$ | Inference Time (ms) | Parameters (M) | FLOPs (G) | DOTA mAP | DOTA CorLoc | DIOR mAP | DIOR CorLoc |
|---|---|---|---|---|---|---|---|
| PCIR [3] | 22.4 | 138 | 147.4 | 26.16 | 48.41 | 24.92 | 46.12 |
| 1 | 17.0 | 159 | 130.2 | 27.80 | 49.75 | 27.15 | 47.64 |
| 2 | 23.1 | 160 | 132.5 | 28.73 | 53.26 | 28.07 | 51.01 |
| 3 | 25.8 | 162 | 134.8 | 30.06 | 55.77 | 29.36 | 53.42 |
| 4 | 33.5 | 166 | 137.1 | 30.10 | 55.88 | 29.39 | 53.41 |

optimal setting $S$ = 8, the number and quality of object patches generated by the SPG are sufficient to cover diverse objects without redundancy. The coarse grid ($S$ = 4) causes 48% fewer patches than that of $S$ = 8 because it has limited ability to extract object edge information, causing a large number of false negatives. Compared with $S$ = 8, $S$ = 16 introduces more noisy proposals, resulting in an overall decrease of the mAP by about 4% on the two datasets. In conclusion, $S$ = 8 balances coverage and precision, which is dense enough to capture objects, even the tiny ones.

As shown in Fig. 5(b), our SCIR achieves a peak 72.8% Mask IoU and 47.7% recall with Top-K = 15, balancing the quality and consistency of SAM-generated mask proposals. The quantitative results reveal a critical inflection point at Top-K = 15, beyond which performance gains are limited but the computational burden continues to grow. Higher Top-K introduces more low-confidence peaks like background clutter in cross attention maps, causing the SAM to generate fragmented or false-positive masks. This demonstrates diminishing returns for Top-K >15, where each additional prompt yields less than 0.1% recall gain per 1-ms time increase.

Comprehensive parameter ablation experiments show that the parameter combination of $S$ = 8 and Top-K = 15 is optimal for our SCIR, which reveals critical performance–efficiency tradeoffs.

*3) Ablation for Different Cascade Levels in CPIR:* To evaluate the performance sensitivity to different cascade levels in CPIR's refinement branches, we conduct the ablation experiment of $K$ values in Eq. (9). As shown in Table XII, $K$ = 3, respectively, achieves the mAP of 30.06% and 29.36% on the DOTA [61] and DIOR [62] datasets, outperforming $K$ = 1 by 2.26% and 2.19% and $K$ = 2 by 1.33% and 1.31%, which confirms that 3-stage refinement sufficiently consolidates contextual cues without over-processing. Compared with $K$ = 3, $K$ = 4 significantly increases the inference time from 25.8 to 33.5 ms, while only improving the mAP by 0.04% and 0.03% on the two datasets. High-level like $K$ = 4 makes the models over-refinement, erroneously merging adjacent objects. Simultaneously, the redundant computations inherent in such high-level exacerbate overfitting and degrade throughput without meaningful accuracy gains. Comprehensive ablation experiment of cascade levels shows that $K$ = 3 is optimal for different datasets or object categories in our SCIR, which strikes a balance between performance and efficiency.

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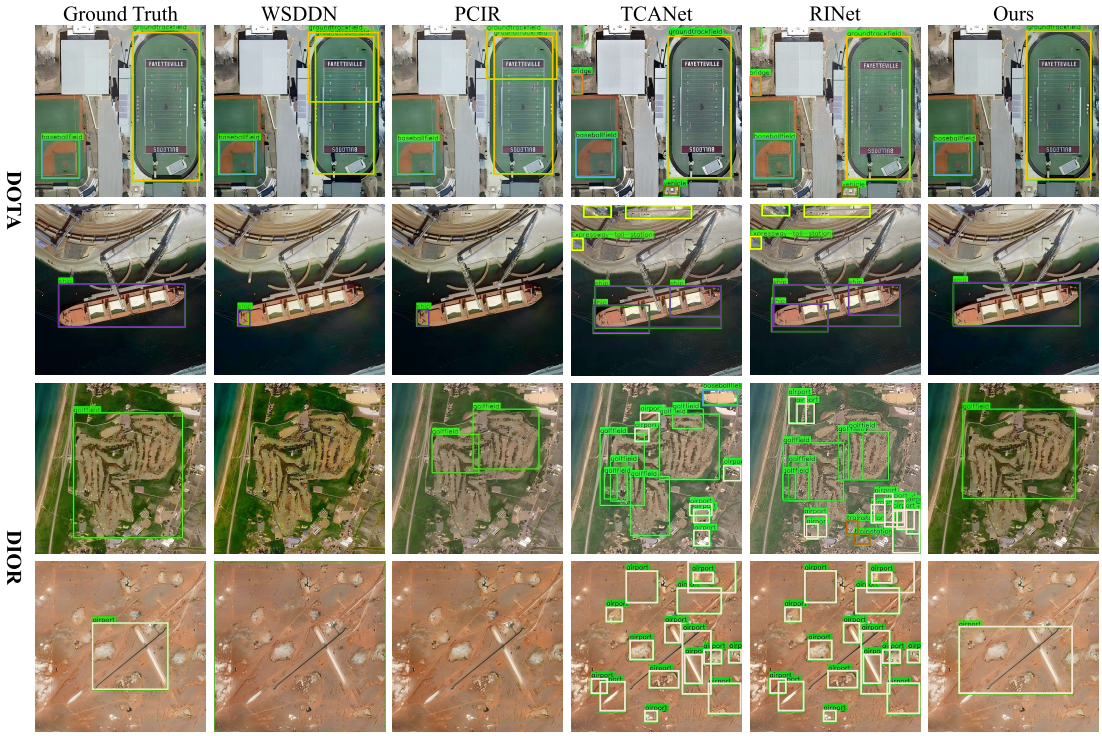



Fig. 6. Four advanced WSOD methods are subjectively compared with our SCIR on the DOTA and DIOR datasets.

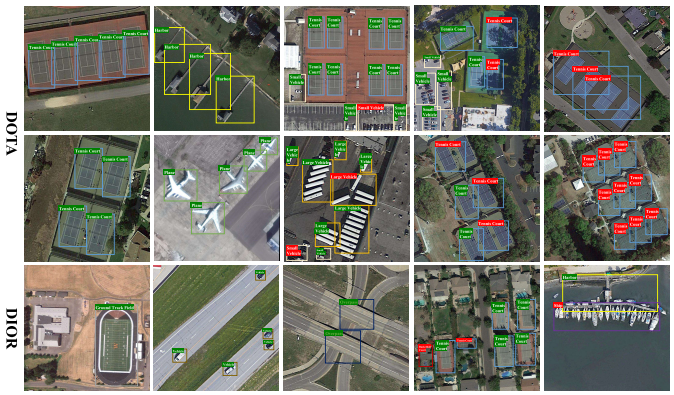



Fig. 7. Some qualitative detection results on the DOTA (30.06% mAP) and DIOR (29.36% mAP) testing split. Words of the annotation category with the green background and the red one, respectively, indicate success and failure cases.

### *E. Visualization Analysis*

Fig. 6 presents the visual comparison results obtained from the DOTA [61] and DIOR [62] datasets, where objects are delineated with green bounding boxes. Additionally, the category of each object is labeled alongside its corresponding bounding box. Compared with four SOTA methods [3], [6], [7], [12], our approach achieves more precise localization and more accurate identification of objects.

Fig. 7 shows the qualitative visualization of successful and failed instances on the two datasets. It is evident that our SCIR has a better performance for detecting small objects in multiscale problems to a certain extent. However, SCIR struggles with objects under large rotation angles or blur, as exemplified by the tennis court case in Fig. 7. Specifically, when the tennis court is rotated beyond 30 degrees or appears blurred due to sensor motion, the SPG fails to capture complete object structures, leading to missed or fragmented detections. This occurs because extreme rotations and blur degrade feature distinctiveness, causing both SAM's mask predictions and CPIR to lose precision. Moreover, the harbor with high resolution in Fig. 6 and the tennis court with haze condition in Fig. 7 demonstrate that the detection performance of our SCIR is susceptible to significant disruption when confronted with extreme weather and high-resolution edge cases.

Fig. 8 shows the qualitative visualization of mask maps and CAM maps in the SPG, and feature attention maps for three modules on the two datasets. The mask maps demonstrate strong segmentation capabilities for well-defined objects but struggle with fragmentation in complex or dense scenarios. CAM maps provide reliable class-specific localization for prominent features but often suffer from partial activation for smaller or low-contrast objects. In the category of plane on the DOTA [61] dataset, feature maps in the CEFE show that it effectively distinguishes densely arranged objects, mitigating the partial dominance issue. Compared to the mask maps and CAM maps in the SPG, these feature maps processed by the CEFE provide more uniform coverage across the entire object, reducing background interference. What is more, feature attention maps for three modules, especially CPIR, effectively highlight salient object regions yet exhibit sensitivity to background clutter and occlusion.

Although our SCIR has problems in dealing with objects of large rotation angles and scene blur objects, take the tennis court as an example, the proposals are detected well when their object rotation angle is small, and the gap with the

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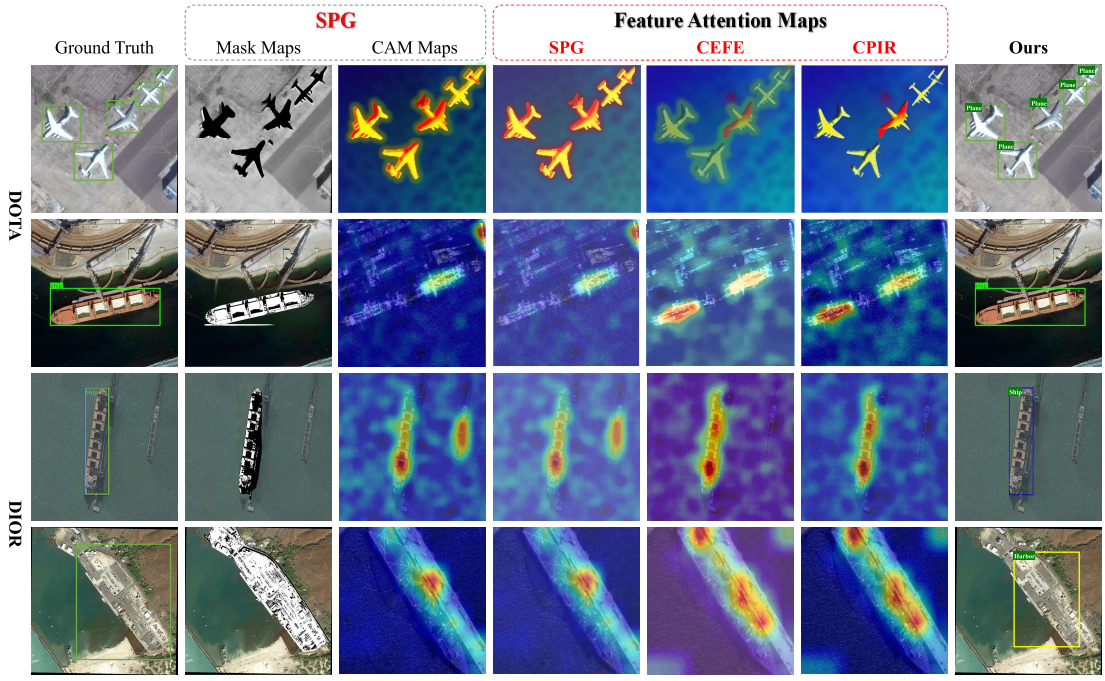



Fig. 8. Visualization of mask maps and CAM maps in the SPG, and feature attention maps for three modules on the DOTA and DIOR datasets.

background features is large. By leveraging weak supervision of image-level labels rather than bounding boxes, our SCIR democratizes access to critical applications like environmental monitoring and disaster response. For instance, SAM-generated proposals in the SPG enable high-recall detection of multiscale objects, including tiny or densely clustered elements, facilitating rapid assessment of deforestation, urbanization, or postdisaster damage without exhaustive human effort.

According to the visualization results, comprehensive experiments show that the proposed SCIR effectively alleviates the issues of low-quality proposals and RSI interference.

## V. Conclusion

In this article, we introduce a novel SCIR method for WSOD. We first design a CEFE module to activate the entire object's features by capturing both the global context of the visual scene and the context-guided feature pyramid. The SPG represents a groundbreaking integration of the SAM with WSL, addressing the challenges of low-quality and redundant WSOD proposals. Based on the WSOD framework, CPIR refocuses the detection network's attention from distinctive local parts to the entire object and other potential instances by utilizing both local and global context information. Moreover, we combine enhanced contextual instance refinement with regression supervision in the regression branch to dynamically correct the object detection frame more precisely. Extensive experiments on the DOTA and DIOR datasets clearly demonstrate that the proposed method is quite effective in addressing the multiscale object issues and achieves SOTA performance, which means that our SCIR significantly makes improvements in object detection accuracy over most existing weakly supervised approaches.

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